

Graphical Excellence Report

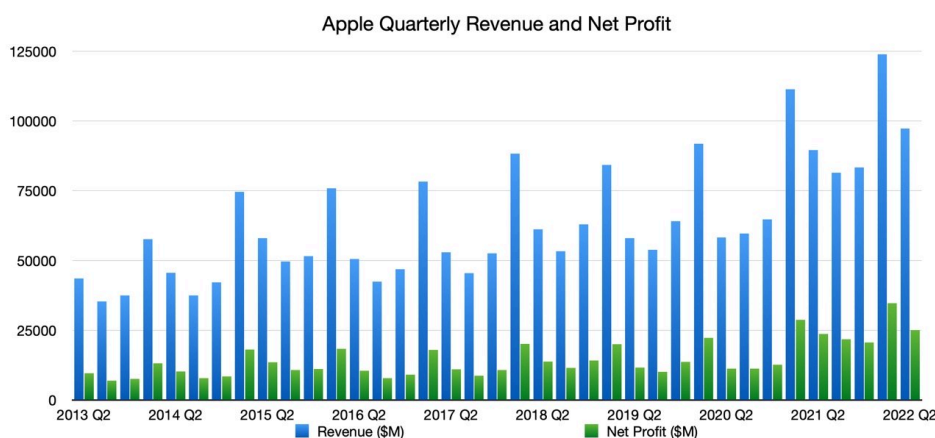
DATA 607

Fall 2025

1 Introduction

Data visualizations are powerful tools that help us understand complex information quickly and intuitively. A well-designed graph can reveal trends, patterns, and insights that might be hard to spot in raw numbers or tables, while a poorly designed one can mislead or confuse the audience. In this report, we examine four different visualizations to understand what makes a graph effective or ineffective. Two of the graphs—Apple's iPhone quarterly revenue chart and Tesla's vehicle delivery chart—clearly communicate trends over time and make comparisons easy. The other two—the Obama–Clinton decision tree and the “Ebb and Flow of Movies” box office visualization—illustrate common pitfalls in design, such as oversimplifying complex data or creating visual clutter. By analyzing these examples, this report highlights the principles of graphical excellence and demonstrates how thoughtful visualization can enhance understanding and decision-making.

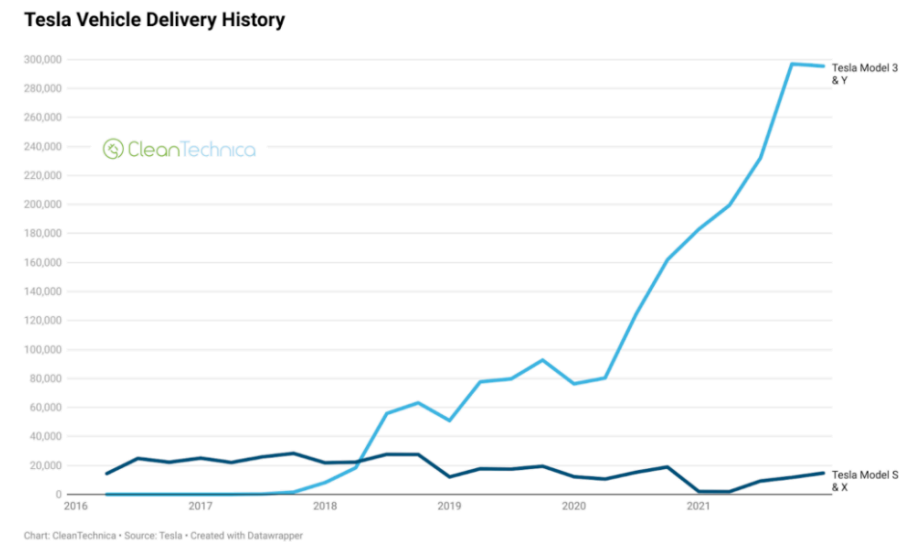
1.0.1 Effective Graph Example 1: Apple's iPhone quarterly revenue chart



Apple's iPhone quarterly revenue chart, from The CRO Club (croclub.com: <https://croclub.com/data-reporting/data-visualization-real-life-examples/>)

Makes use of vertical bars to show how much revenue Apple earned in each quarter, with taller bars representing higher sales. This makes it easy to instantly see trends—for example, the chart clearly shows that the fourth quarter of each year has the tallest bars, reflecting the holiday season when iPhone sales peak. You don't need to read every number to understand that some quarters perform much better than others. It also makes comparisons simple: you can quickly see how a slow summer quarter stacks up against a busy winter quarter. Think of it like a chart of students' heights: even without knowing the exact numbers, you can tell who is tallest. The chart is effective because it combines simplicity with clarity—it not only shows revenue figures but also highlights seasonal patterns and sales trends over time, helping anyone, even without a business background, understand Apple's sales performance and the impact of seasonal demand.

1.0.2 Effective Graph Example 2: Tesla vehicle delivery chart



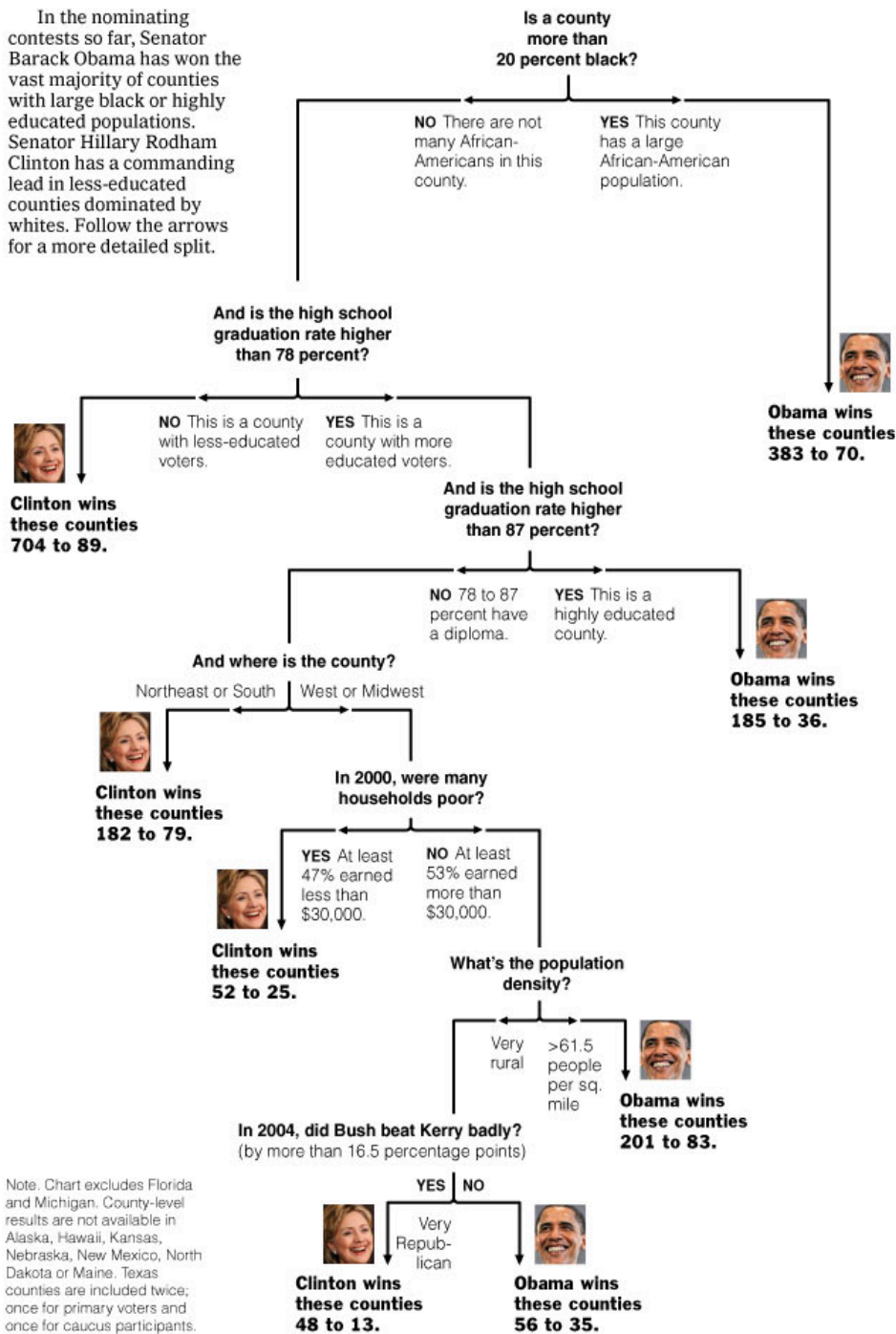
Tesla's vehicle delivery chart, from The CRO Club ([croclub.com](https://croclub.com/data-reporting/data-visualization-real-life-examples/): <https://croclub.com/data-reporting/data-visualization-real-life-examples/>),

uses a line graph to show the number of cars Tesla delivered each quarter over time. The line rises and falls to represent increases and decreases in deliveries, making trends immediately visible without needing to read every number. For example, the chart clearly shows a sharp spike in deliveries in the fourth quarter of 2022, highlighting Tesla's strong year-end push to meet production targets and market demand. By following the direction of the line, you can also compare performance across quarters or see growth patterns over multiple years, which helps illustrate the company's long-term expansion. Think of it like tracking your own bank balance over time: even without knowing the exact figures, you can see when your savings grew faster or slowed down. The chart is effective because it combines simplicity with storytelling: it doesn't just show numbers—it visually communicates Tesla's growth, seasonal effects, and overall trajectory, making it easy for anyone, even without a business background, to understand the company's delivery trends and performance over time.

1.0.3 Ineffective Example 1: Obama–Clinton Decision Tree

Decision Tree: The Obama-Clinton Divide

In the nominating contests so far, Senator Barack Obama has won the vast majority of counties with large black or highly educated populations. Senator Hillary Rodham Clinton has a commanding lead in less-educated counties dominated by whites. Follow the arrows for a more detailed split.



"Showing the Obama–Clinton Divide in Decision Tree Infographic" (2008). Link:

<https://flowingdata.com/2008/04/23/showing-the-obama-clinton-divide-in-decision-tree-infographic/>
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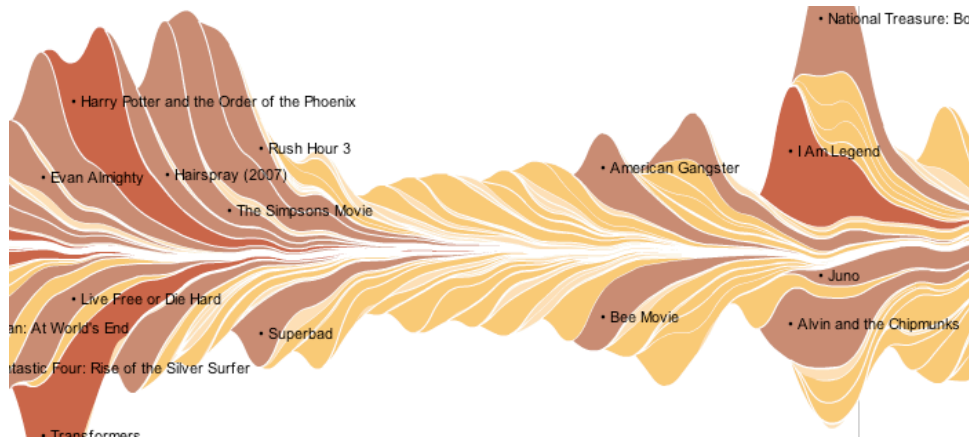
During the 2008 Democratic primaries, The New York Times published a fancy chart that looked like a tree, trying to explain why some counties voted for Obama and others for Clinton. The idea was to show that things like race, income, and education could predict how people would vote.

But here's the problem: this chart makes voting seem too simple. It's like saying, "If you're X years old and earn Y dollars, you must vote for Obama." In real life, people's political choices are messy—many factors overlap, and there's a lot of uncertainty. Could a county with mostly college graduates and moderate incomes still vote for Clinton? Absolutely. Can someone's personal experiences or local issues sway their vote? Of course.

On top of that, the chart included candidate photos and extra decorations that don't actually tell us anything about the data—they just make it look cooler. As statisticians Gelman and Unwin point out, this kind of visualization doesn't honestly summarize what's really going on. It gives the illusion that we can neatly classify voters, when in reality, politics is much more complicated.

So next time you see a chart that claims it can “predict” people's choices, ask yourself: “Is this really showing reality, or is it just a simplified story that looks neat?”

1.0.4 Ineffective Example 2: The Ebb and Flow of Movies: Box Office Receipts, 1986–2008.



Byron & Wattenberg (2008). “Stacked Graphs—Geometry & Aesthetics.” IEEE Transactions on Visualization and Computer Graphics.

Popularized in The New York Times Interactive: “The Ebb and Flow of Movies: Box Office Receipts, 1986–2008.” Archive link (via FlowingData): <https://flowingdata.com/2008/02/25/ebb-and-flow-of-box-office-receipts-over-past-20-years/> (<https://flowingdata.com/2008/02/25/ebb-and-flow-of-box-office-receipts-over-past-20-years/>)

The New York Times made an interactive graphic called “The Ebb and Flow of Movies” to show U.S. box office earnings over 20 years. Each movie is drawn as a colorful, wavy band: the thicker the band, the more money the movie made, and the longer it stretches, the longer it stayed in theaters. The chart looks artistic, almost like layers of rivers flowing together. But it's hard to use if you actually want to answer questions about the data. For example, if someone asked:

“Which movie made the most money in 2005?”

or “How did Titanic compare to other movies when it was released?”

the wavy shapes make it tricky to tell. The bands overlap and bend so much that you can't easily compare one movie's size to another. The colors blend together too, which adds to the difficulty.

So while the streamgraph is fun to look at, it doesn't do a good job of clearly showing which movies earned more or less money.

2 Conclusion

visualizations can be incredibly powerful tools for understanding complex data, but their effectiveness depends on clarity, accuracy, and the way information is presented. The Tesla vehicle delivery line graph and Apple iPhone quarterly revenue column chart are prime examples of effective visualizations: they clearly show trends over time, allow easy comparisons, and highlight meaningful patterns such as seasonal sales spikes or year-over-year growth. Readers can grasp the key insights at a glance without needing to interpret raw numbers, making the data both accessible and engaging. In contrast, the “Ebb and Flow of Movies: Box Office Receipts, 1986–2008” graph and the Obama–Clinton decision tree demonstrate the pitfalls of ineffective visualization.

The movie revenue graph's complexity and clutter make it difficult to interpret, while the decision tree oversimplifies political behavior and distracts with unnecessary design elements, giving a misleading sense of precision. Together, these examples illustrate that the best visualizations communicate data clearly, highlight meaningful trends, and make complex information intuitive, whereas poorly designed graphics can confuse or mislead the audience.